

Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

Yichao Jin, Ph.D.

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1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S14 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering reflects the endpoint definitions used in the clinical trials: DMFS counts either distant metastasis *or death* as an event, so the probability of remaining alive and distant-metastasis-free cannot exceed overall survival ($DMFS \leq OS$). Similarly, RFS counts recurrence *or death*, so $RFS \leq DMFS$. The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic



Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [4]. At earlier time points, the 18-month RFS rates from the primary analysis [5] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

3 Clinical Input Provenance

Table S3 summarizes the provenance of all clinical inputs used in the model. The evidence status column classifies each input into one of five categories:

- **Published:** The value is directly reported in the cited source.
- **Digitized:** The value was extracted from a published Kaplan–Meier curve using Web-PlotDigitizer v4.7.
- **Derived:** The value is mathematically derived from other inputs (e.g., ratio of survival functions).
- **Modeled:** The value is produced by the model’s parametric extrapolation (e.g., long-term survival beyond observed data).
- **Topline only:** The source reports a statistically significant result but the numerical hazard ratio or landmark survival probabilities have not yet been disclosed.

Table S3: Clinical Input Provenance

Input	Value	Evidence Status	Source
Input	Value	Evidence Status	Source
<i>KEYNOTE-942 (Phase 2b, 5-year follow-up)</i>			
Recurrence-Free Survival (RFS)			
18-month RFS, combo	80.4%	Digitized	Fig. 1, Khattak 2026
24-month RFS, combo	78.3%	Digitized	Fig. 1, Khattak 2026
36-month RFS, combo	73.8%	Digitized	Fig. 1, Khattak 2026
48-month RFS, combo	72.4%	Digitized	Fig. 1, Khattak 2026
60-month RFS, combo	68.8%	Published	Khattak 2026
60-month RFS, combo 95% CI	56.3–78.3%	Published	Khattak 2026
18-month RFS, pembro	62.2%	Digitized	Fig. 1, Khattak 2026
24-month RFS, pembro	60.0%	Digitized	Fig. 1, Khattak 2026
36-month RFS, pembro	55.6%	Digitized	Fig. 1, Khattak 2026
48-month RFS, pembro	49.1%	Digitized	Fig. 1, Khattak 2026
60-month RFS, pembro	49.1%	Published	Khattak 2026
60-month RFS, pembro 95% CI	33.3–63.0%	Published	Khattak 2026
Distant Metastasis-Free Survival (DMFS)			
18-month DMFS, combo	92.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, combo	90.8%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, combo	85.6%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, combo	83.9%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, combo	–	Not reported	N/A
18-month DMFS, pembro	73.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, pembro	70.6%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, pembro	–	Not reported	N/A
Overall Survival (OS)			

Input	Value	Evidence Status	Source
18-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
24-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
36-month OS, combo	93.8%	Digitized	Fig. 1, Khattak 2026
48-month OS, combo	92.2%	Published	Khattak 2026
60-month OS, combo	92.2%	Published	Khattak 2026
18-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
24-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
36-month OS, pembro	85.6%	Digitized	Fig. 1, Khattak 2026
48-month OS, pembro	85.6%	Published	Khattak 2026
60-month OS, pembro	71.3%	Published	Khattak 2026
Hazard Ratios			
RFS HR (combo vs. pembro)	0.510	Published	Khattak 2026
RFS HR 95% CI	0.294–0.887	Published	Khattak 2026
DMFS HR (combo vs. pembro)	0.411	Published	Khattak 2026
DMFS HR 95% CI	0.200–0.843	Published	Khattak 2026
OS HR (combo vs. pembro)	0.471	Published	Khattak 2026
OS HR 95% CI	0.165–1.345	Published	Khattak 2026
OS events	14 (7 per arm)	Published	Khattak 2026
<i>INTerpath-001 (Phase 3, topline announcement)</i>			
RFS	Statistically significant	Topline only	Merck/Moderna 2026
DMFS	Statistically significant	Topline only	Merck/Moderna 2026
Numerical HR / KM curves	Not yet disclosed	Topline only	N/A
<i>Modeled / Derived Values</i>			
Fitted log-normal parameters			
OS μ (combo), σ (combo)	8.396, 3.096	Derived	Least-squares fit
OS μ (pembro), σ (pembro)	4.841, 1.131	Derived	Least-squares fit
r_DM μ (combo), σ (combo)	7.113, 2.480	Derived	Ratio fit
r_DM μ (pembro), σ (pembro)	7.042, 5.000	Derived	Ratio fit
r_LR μ (combo), σ (combo)	8.874, 5.000	Derived	Ratio fit
r_LR μ (pembro), σ (pembro)	5.913, 2.674	Derived	Ratio fit
Long-term extrapolation			
10-year OS, combo	84.1%	Modeled	Log-normal extrapolation
10-year OS, pembro	51.6%	Modeled	Log-normal extrapolation
20-year OS, combo (GP-constrained)	60.1%	Modeled	GP hazard floor
20-year OS, pembro (GP-constrained)	28.5%	Modeled	GP hazard floor
30-year OS, combo (GP-constrained)	22.2%	Modeled	GP hazard floor
30-year OS, pembro (GP-constrained)	10.5%	Modeled	GP hazard floor
40-year OS, combo (GP-constrained)	2.0%	Modeled	GP hazard floor

3.1 Observed vs. Fitted Survival Curves



Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

4 Survival Model Selection

4.1 Component-Level Parametric Fit

Table S4: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

4.2 Joint Model Selection

A total of 15,625 distribution combinations (5 distributions x 6 curves) were evaluated for structural validity at 40 years. All 15,625 satisfied the $RFS \leq DMFS \leq OS$ ordering constraint (0 crossing violations). The full results are available online ([output/survival_combos_15625.csv](#)).[†] AIC is reported as an exploratory fit diagnostic; landmark probabilities from the same Kaplan–Meier curve are serially

correlated, so standard AIC interpretation (which assumes independent observations) does not strictly apply.

Table S5: Top 10 Joint Model Combinations by Total AIC¹ (5 distributions, 15,625 combinations)

Rank	OS	r _{DM}	r _{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / GG	LL / W	-224.76	76.3%	0.0%
2	LN / W	LN / GG	LL / LL	-224.67	76.3%	0.0%
3	LN / W	Exp / GG	LL / W	-224.65	76.3%	0.0%
4	LLog / W	LN / GG	LL / W	-224.57	72.9%	0.0%
5	LN / W	LN / GG	LN / W	-224.55	76.3%	0.0%
6	LN / W	LN / GG	LL / LN	-224.53	76.3%	0.0%
7	LN / W	Exp / GG	LL / LL	-224.53	76.3%	0.0%
8	W / W	LN / GG	LL / W	-224.52	71.1%	0.0%
9	LLog / W	LN / GG	LL / LL	-224.48	72.9%	0.0%
10	LN / W	LN / GG	LN / LL	-224.46	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential, GG = generalized gamma. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3% under unconstrained extrapolation, substantially higher than GP survival at age 101 (2.0%). This motivated the GP mortality hazard floor (Section 5.1).

4.3 Observed versus Fitted Landmark Probabilities

Table S6 reports the base-case log-normal model predictions at each KEYNOTE-942 landmark time point, together with residuals (fitted minus observed). The mean absolute error (MAE) across all 27 landmark points is 1.98 percentage points, and the root mean squared error (RMSE) is 2.79 percentage points.

The largest residual is at the pembrolizumab-arm RFS 60-month landmark (observed 49.1%, fitted 40.9%, residual -8.2 percentage points). This reflects two structural features of the model: (i) the ratio-based construction fits $r_{LR} = \text{RFS}/\text{DMFS}$ as a separate parametric curve, so residuals at the RFS level compound errors from both the DMFS and r_{LR} components; and (ii) the observed pembrolizumab RFS is identical at 48 and 60 months (both 49.1%), creating a flat plateau that no standard parametric survival distribution can reproduce—this plateau may reflect sparse event accumulation and sampling variability in the small pembrolizumab arm ($n = 52$). The DMFS pembrolizumab 48-month landmark also shows a notable residual (-5.6 percentage points) for the same reason.

Despite these local deviations, the model provides a reasonable parametric representation of the overall survival trajectory: 22 of 27 landmarks have absolute residuals below 3 percentage points, and the combo-arm curves (which drive the incremental survival benefit and hence the ICER) are well-fitted across all time points (maximum absolute residual 1.5 percentage points for RFS, 1.4 percentage points for DMFS, 0.7 percentage points for OS). The extrapolated long-term survival predictions are consistent with external trial evidence (Section 6) and general population mortality constraints (Section 5.1).

¹AIC is reported as an exploratory fit diagnostic. Landmark probabilities derived from the same Kaplan–Meier curve are serially correlated (each time point depends on the same censoring history), so the standard AIC assumption of independent observations does not strictly apply. Model selection was primarily guided by RMSE, visual fit, and clinical plausibility.

Table S6: Observed versus Fitted Landmark Survival Probabilities

Endpoint	Arm	Month	Observed	Fitted	Residual	Residual	Rel. Error
OS	Combo	18	96.0%	96.4%	+0.4%	0.4%	0.4%
OS	Combo	24	96.0%	95.5%	-0.5%	0.5%	0.5%
OS	Combo	36	93.8%	94.1%	+0.3%	0.3%	0.3%
OS	Combo	48	92.2%	92.9%	+0.7%	0.7%	0.8%
OS	Combo	60	92.2%	91.8%	-0.4%	0.4%	0.4%
OS	Pembro	18	93.4%	96.2%	+2.8%	2.8%	3.0%
OS	Pembro	24	93.4%	93.4%	+0.0%	0.0%	0.0%
OS	Pembro	36	85.6%	87.2%	+1.6%	1.6%	1.9%
OS	Pembro	48	85.6%	81.0%	-4.6%	4.6%	5.4%
OS	Pembro	60	71.3%	75.0%	+3.7%	3.7%	5.2%
DMFS	Combo	18	92.0%	92.3%	+0.3%	0.3%	0.3%
DMFS	Combo	24	90.8%	90.3%	-0.5%	0.5%	0.5%
DMFS	Combo	36	85.6%	87.0%	+1.4%	1.4%	1.6%
DMFS	Combo	48	83.9%	84.1%	+0.2%	0.2%	0.3%
DMFS	Pembro	18	73.0%	77.0%	+4.0%	4.0%	5.4%
DMFS	Pembro	24	70.6%	73.1%	+2.5%	2.5%	3.6%
DMFS	Pembro	36	65.4%	66.0%	+0.6%	0.6%	1.0%
DMFS	Pembro	48	65.4%	59.8%	-5.6%	5.6%	8.6%
RFS	Combo	18	80.4%	81.8%	+1.4%	1.4%	1.8%
RFS	Combo	24	78.3%	79.0%	+0.7%	0.7%	0.9%
RFS	Combo	36	73.8%	74.5%	+0.7%	0.7%	0.9%
RFS	Combo	48	72.4%	70.9%	-1.5%	1.5%	2.1%
RFS	Combo	60	68.8%	67.9%	-0.9%	0.9%	1.3%
RFS	Pembro	18	62.2%	67.4%	+5.2%	5.2%	8.3%
RFS	Pembro	24	60.0%	62.2%	+2.2%	2.2%	3.6%
RFS	Pembro	36	55.6%	53.6%	-2.0%	2.0%	3.7%
RFS	Pembro	48	49.1%	46.6%	-2.5%	2.5%	5.1%
RFS	Pembro	60	49.1%	40.9%	-8.2%	8.2%	16.7%
			MAE		1.98 percentage points		
			RMSE		2.79 percentage points		
			Max Residual 		8.2 percentage points (RFS, pembrolizumab, 60 mo)		

Note: Fitted values are from the base-case log-normal partitioned survival model at observed KEYNOTE-942 landmark time points. Residuals are fitted minus observed; positive values indicate model overprediction.

4.4 Lifetime OS Extrapolation by Distribution



Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four conventional distributions produce wildly divergent 40-year survival estimates (0% to 76.3%). Generalized gamma was included in the formal model-selection analysis but is not shown here for visual clarity.

5 General-Population Mortality and Structural Validation

5.1 GP Mortality Implementation

The GP mortality hazard floor is applied to modeled OS from month 60 onward.

Life table source: US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:** Annual life table survival linearly interpolated to monthly resolution; monthly hazards derived as $h_{GP}(t) = -\ln[S_{GP}(t+1)/S_{GP}(t)]$. **Constraint formula (mortality hazard floor):** from month 60 onward, the model mortality hazard is floored at the age- and sex-matched general-population hazard:

$$h_{\text{model}}^{\text{con}}(t) = \max[h_{\text{param}}(t), h_{GP}(t)] \quad t \geq 60 \text{ months}, \quad (1)$$

where $h_{\text{param}}(t) = -\ln[S_{\text{param}}(t+1)/S_{\text{param}}(t)]$ is the monthly hazard implied by the fitted parametric survival curve. The constrained survival is reconstructed as $S_{\text{model}}^{\text{con}}(t) = \exp[-\sum_0^t h^{\text{con}}(u)]$. This is a **hazard floor**, not a survival cap: a cohort that has already experienced elevated (cancer-related) mortality in the first 5 years remains below general-population survival at every subsequent time point, whereas a survival cap ($S = \min[S_{\text{param}}, S_{GP}]$) would incorrectly force the two curves to coincide whenever the parametric tail exceeded the general-population level. Both arms are subject to the same background mortality floor.

5.2 Long-Term Survival Diagnostics

Table S7: Long-Term Survival Predictions vs. General Population. GP survival is conditional on being alive at baseline age 61 and is calculated as $S_{GP}(t) = l_{61+t}/l_{61}$ using the sex-weighted 2019 US life tables (64% male, 36% female); the terminal 100+ life-table value is used for the 40-year row.

Years from baseline	Attained age	Combo OS	Pembro OS	GP survival
5	66	91.6%	74.1%	94.1%
10	71	84.1%	51.6%	86.4%
15	76	74.0%	37.6%	76.1%
20	81	60.1%	28.5%	61.9%
30	91	22.2%	10.5%	23.1%
40	101	2.0%	0.9%	2.0%

5.3 OS versus General-Population Survival



GP-constrained curves (solid) use a mortality hazard floor from month 60. The constrained combo OS stays below the GP curve at all time points, reflecting the excess mortality experienced in the first 5 years. Unconstrained log-normal (dotted) produces implausibly high 40-year survival (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model (hazard floor from 60 months). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61. The constrained combo curve remains below the GP curve at all time points, reflecting the hazard-floor property: a cohort with prior excess mortality does not "catch up" to general-population survival.

5.4 Structural Quality Assurance

Table S8: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0

6 External Plausibility Assessment

6.1 External Plausibility Assessment Landmarks

Table S9: External Plausibility Assessment of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	—	51% [6]
RFS	7	31.9%	50% [7]	46% [6]
DMFS	5	53.8%	—	59% [6]
DMFS	7	44.9%	54% [7]	55% [6]
OS	5	74.5%	NR	76% [8]
OS	7	64.2%	NR	71% [6]

NR: Not reported. CheckMate 238 evaluated nivolumab (not pembrolizumab). The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population (stage IIIB–IV including resected stage IV melanoma vs IIIA–C in KEYNOTE-054 and IIIB–IV in CheckMate 238). Only published landmark values are included; no extrapolated estimates are used.[6, 7]

6.2 Longitudinal External Plausibility Assessment



Figure S5: External plausibility assessment of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942 observed data. Squares: KEYNOTE-054 (pembrolizumab, 7-year). Triangles: CheckMate 238 (nivolumab, 5- and 7-year). Only published landmark values are shown.

7 Model Inputs

7.1 Complete Parameter Table

Table S10: Full Model Parameter Inventory. *Note:* DSA ranges are based on fixed percentage perturbations ($\pm 20\%$ or $\pm 40\%$) of the base-case fitted values to explore parameter sensitivity. PSA distributions use Normal sampling with standard errors derived from the nonlinear least-squares Jacobian covariance, reflecting curve-fit uncertainty rather than the full sampling uncertainty of the original censored trial data.

Parameter	Base	DSA Low	DSA High	PSA Distribution	Source
<i>Clinical Parameters</i>					
OS μ – combo	8.396	5.038 (−40%)	11.754 (+40%)	Normal (SE from Jacobian)	Fitted
OS σ – combo	3.096	1.858 (−40%)	4.334 (+40%)	Normal (SE from Jacobian)	Fitted
OS μ – pembro	4.841	3.873 (−20%)	5.809 (+20%)	Normal (SE from Jacobian)	Fitted
OS σ – pembro	1.131	0.905 (−20%)	1.357 (+20%)	Normal (SE from Jacobian)	Fitted
r _{DM} μ – combo	7.113	4.268 (−40%)	9.958 (+40%)	Normal (SE from Jacobian)	Fitted
r _{DM} σ – combo	2.480	1.488 (−40%)	3.472 (+40%)	Normal (SE from Jacobian)	Fitted
r _{DM} μ – pembro	7.042	5.634 (−20%)	8.450 (+20%)	Normal (SE from Jacobian)	Fitted
r _{DM} σ – pembro	5.000 [†]	3.000 (−40%)	7.000 (+40%)	Normal (SE from Jacobian)	Fitted
r _{LR} μ – combo	8.874	5.324 (−40%)	12.424 (+40%)	Normal (SE from Jacobian)	Fitted
r _{LR} σ – combo	5.000 [†]	3.000 (−40%)	7.000 (+40%)	Normal (SE from Jacobian)	Fitted
r _{LR} μ – pembro	5.913	4.730 (−20%)	7.096 (+20%)	Normal (SE from Jacobian)	Fitted
r _{LR} σ – pembro	2.674	1.604 (−40%)	3.744 (+40%)	Normal (SE from Jacobian)	Fitted
Treatment duration (pembro)	18 cycles (1 yr)	9	24	Fixed	[4]
Treatment duration (intismeran)	9 doses (27 weeks)	6	12	Fixed	[4]
Baseline age	61 years	55	67	Fixed	[4]
Male proportion	64%	–	–	Fixed	[4]
<i>Costs (2026 USD)</i>					
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Fixed (not sampled)	Assumed
Pembrolizumab (annual)	\$220,896	\$176,717	\$265,075	Gamma (CV=0.2)	WAC (CMS)
Tumor sequencing	\$1,000	\$800	\$1,200	Fixed (DSA only)	CMS
Administration — pembrolizumab IV infusion (CPT 96413)	\$119.29	–	–	Fixed (not sampled; symmetric across arms)	CMS
Administration — intismeran IM injection (CPT 96372, proxy code)	\$13.91	\$0	\$120	Fixed (DSA only)	CMS
AE management (one-time)	\$15,000	\$12,000	\$18,000	Fixed (DSA only)	[3]
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.2)	[3]
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.2)	[1]
<i>Utilities</i>					
RF state	0.83	0.80	0.86	Beta (SE=0.03)	[2]
LR state	0.64	0.61	0.67	Beta (SE=0.03)	[2]
DM state	0.55	0.52	0.58	Beta (SE=0.03)	[2]
<i>Economic Parameters</i>					
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)	[9]
Time horizon	40 years	10	20	Fixed (scenario)	Assumed
Willingness-to-pay threshold	\$100K–\$150K/QALY	\$50K	\$200K	Fixed (scenario)	[10]
Cost base year	2026 USD	–	–	–	Assumed
GP mortality source	US 2019 Life Tables	–	–	–	NVSS (Arias 2022)

[†] Boundary solution: σ reaches the upper bound of the estimation window (5.0); the ratio curve is approximately constant. Jacobian covariance is approximate.

8 Additional Results

8.1 Lifetime Cost Decomposition

Table S11: Lifetime Cost Decomposition by Component (GP-constrained, 40-year horizon)

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,931	\$217,931	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$2,241	\$2,116	\$124
Adverse events	\$15,000	\$0	\$15,000
LR management	\$93,867	\$73,340	\$20,527
DM management	\$407,465	\$446,762	−39,298
Total	\$937,504	\$740,150	\$197,354

8.2 LY and QALY Decomposition

Table S12: Life-Year and QALY Decomposition by Health State

Outcome	Combo	Pembro	Incremental
Life-years (undiscounted)	24.39	14.71	9.68
Life-years (discounted 3%)	16.30	10.66	5.64
RF state QALYs	9.018	4.580	4.438
LR state QALYs	1.669	1.304	0.365
DM state QALYs	1.556	1.706	−0.150
Total QALYs	12.243	7.590	4.653

Health benefit is driven by RF state QALYs (4.438 gained), with a modest additional contribution from LR (0.365) and a small reduction in DM state QALYs (−0.150) under the general-population mortality constraint.

9 Uncertainty and Structural Scenario Analyses

9.1 Deterministic Sensitivity Analysis

Table S13: One-Way Deterministic Sensitivity Analysis

Parameter	Base	Low	High	ICER range	Change in NMB
Utility RF	0.83	0.80	0.86	43,926–40,998	\$48,128
Utility LR	0.64	0.61	0.67	42,568–42,257	\$5,132
Utility DM	0.55	0.52	0.58	42,337–42,487	\$2,456
Pembrolizumab annual cost	\$220,896	\$176,717	\$265,075	42,412–42,412	\$0
Intismeran cost	\$200,000	\$160,000	\$240,000	33,816–51,008	\$80,000
LR monthly cost	\$3,000	\$2,400	\$3,600	41,530–43,294	\$8,211
DM monthly cost	\$12,000	\$9,600	\$14,400	44,101–40,723	\$15,719
Discount rate	3%	0%	5%	37,614–48,726	\$522,985
Intismeran IM injection cost	\$13.91	\$0	\$120	42,385–42,616	\$1,073
Tumor sequencing	\$1,000	\$800	\$1,200	42,369–42,455	\$400
AE management	\$15,000	\$12,000	\$18,000	41,767–43,057	\$6,000
OS μ combo	8.396	-40%	+40%	28,283–42,814	\$521,695
OS μ pembro	4.841	-20%	+20%	58,503–Dominant	\$470,922

The maximum finite ICER was \$58,503/QALY across the tested ranges; the combination became dominant when the pembrolizumab OS parameter was increased by 20%.

9.2 Probabilistic Sensitivity Analysis

9.2.1 PSA Implementation Details

PSA with 6,000 Monte Carlo iterations (seed = 42). Survival parameters (μ , σ for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ($r > 0.95$); σ lower-truncated at 0.05. Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Structural constraints (RFS $\leq$ DMFS $\leq$ OS, monotonicity, survival within [0,1]) verified every cycle; all 6,000 draws were structurally valid (0 rejections). No iterations were excluded on the basis of the ICER value.

Note on boundary solutions: Two σ parameters ($r_{\text{DM}}\text{--pembrolizumab}$ and $r_{\text{LR}}\text{--combo}$) reach the upper bound of the estimation window (Table S14). This occurs because the corresponding ratio curves are approximately constant over the observed time points (pembrolizumab r_{DM} : 0.78, 0.76, 0.76, 0.76; combo r_{LR} : 0.87, 0.86, 0.86, 0.86, 0.82), and a log-normal distribution with very large σ approximates a constant function. The Jacobian-based covariance matrices for these two parameter pairs are therefore approximate, as the true Fisher information matrix is not well-defined at a boundary solution. In PSA, samples are drawn from a bivariate normal distribution truncated at the upper bound; this truncation is conservative in that it limits the sampled σ values to the estimation window, preventing extreme tail behavior that could produce structurally invalid survival curves. Sensitivity analyses using alternative distributions (Weibull OS, rank-1 joint specification; Supplementary Table S15) yield ICERs in the range of \$43,000–\$62,000/QALY, suggesting that the cost-effectiveness conclusion is robust to the choice of distribution specification for these components.

Table S14: PSA Parameter Distributions and Sampled Values

Parameter	Distribution	Fitted mean	SE	Correlation	Sampling
OS μ / σ – combo	Bivariate normal	8.396 / 3.096	0.793 / 0.522	0.996	Joint
OS μ / σ – pembro	Bivariate normal	4.841 / 1.131	0.291 / 0.310	0.955	Joint
r _{DM} μ / σ – combo	Bivariate normal	7.113 / 2.480	0.814 / 0.563	0.995	Joint
r _{DM} μ / σ – pembro	Bivariate normal	7.042 / 5.000 [†]	2.278 / 3.130	0.995	Joint
r _{LR} μ / σ – combo	Bivariate normal	8.874 / 5.000 [†]	2.039 / 1.928	0.997	Joint
r _{LR} μ / σ – pembro	Bivariate normal	5.913 / 2.674	0.757 / 0.879	0.984	Joint
Intismeran cost	Fixed	\$200,000	—	—	Fixed (not sampled)
Pembrolizumab annual cost	Gamma (CV=0.2)	\$220,896	—	—	Independent
LR monthly cost	Gamma (CV=0.2)	\$3,000	—	—	Independent
DM monthly cost	Gamma (CV=0.2)	\$12,000	—	—	Independent
Utility RF / LR / DM	Beta (SE=0.03)	0.83 / 0.64 / 0.55	—	—	Independent
Held at base case (not sampled): intismeran acquisition price (price-threshold analysis), tumour sequencing, adverse-event management and the IM injection rate (one-way DSA), and the IV					

[†] Boundary solution: σ reaches the upper bound of the estimation window (5.0); the ratio curve is approximately constant. Jacobian covariance is approximate.

9.2.2 PSA Results

- Mean Δ QALY: 4.95 (95% CI 2.87–7.90).
- Mean Δ Cost: \$163,999 (95% CI –\$677,118 to \$612,309).
- Mean NMB: \$330,809 at \$100K and \$578,213 at \$150K.
- Probability cost-effective: 61.1% at \$50K, 94.8% at \$100K, and 98.5% at \$150K.
- No iterations were excluded based on the ICER; 15.8% of draws were cost-saving, and all 6,000 draws had Δ QALY > 0.

9.2.3 PSA Convergence



Figure S6: PSA convergence across iteration counts (6,000 structurally valid iterations, 0 structural rejections, seed = 42). Cumulative means of Δ QALY, Δ Cost, NMB, and P(CE). Vertical dotted lines at 500, 1,000, 2,000, and 4,000 iterations. Estimates stable from ~500 iterations.

9.2.4 Cost-Effectiveness Plane



Figure S7: CE plane showing 6,000 PSA iterations (blue), WTP lines (\$50K, \$100K, \$150K per QALY), and the deterministic base case (black diamond). 15.8% of iterations fall in the southeast quadrant (cost-saving); all 6,000 iterations had positive Δ QALY. No iterations were excluded based on the ICER value.

9.3 Structural Scenario Analyses

Scenario implementations. Three structural scenarios alter the base-case treatment-effect assumptions. All scenarios preserve the observed 5-year KEYNOTE-942 data unchanged and modify only the extrapolated trajectory beyond 60 months.

OS-survival convergence (5–20 y, Waning V1). On the **survival probability scale**, the incremental OS benefit (combo OS minus pembrolizumab OS) is linearly attenuated from its full base-case value at 60 months to zero at 240 months (20 years). At each monthly cycle t : $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{pembro}}(t) + [S_{\text{combo}}^{\text{base}}(t) - S_{\text{pembro}}(t)] \times \max(0, 1 - \frac{t-60}{180})$. The recurrence (r_{DM}) and local-recurrence (r_{LR}) ratio functions are **not** modified, so the scenario isolates the effect of attenuating the OS benefit while preserving the observed recurrence advantage. The general-population hazard floor continues to apply independently to both arms.

Hazard convergence (5–10 y, Waning V2). On the **cause-specific mortality hazard scale**, the combo arm's monthly all-cause hazard is linearly blended toward the pembrolizumab arm's hazard from month 60 to month 120 (10 years). At each cycle: $h_{\text{combo}}^{\text{waned}}(t) = (1 - w(t))h_{\text{combo}}(t) + w(t)h_{\text{pembro}}(t)$, where $w(t) = \max(0, \min(\frac{t-60}{60}, 1))$. The waned hazard is integrated forward to reconstruct the survival curve: $S_{\text{combo}}^{\text{waned}}(t+1) = S_{\text{combo}}^{\text{waned}}(t) \times \exp(-h_{\text{combo}}^{\text{waned}}(t))$, with $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{combo}}^{\text{base}}(t)$ for $t \leq 60$ months. In this

scenario, r_{DM} and r_{LR} are also linearly converged to pembrolizumab-arm values on the probability scale ($\text{HR} \rightarrow 1$ by 10 years), so that all three survival components reflect complete treatment-effect erosion. The GP hazard floor continues to apply.

No additional OS hazard benefit beyond 5 years. The combination arm follows its own extrapolated OS curve through the 60-month observed trial window, so the observed 5-year OS difference is preserved unchanged; beyond 60 months it takes the pembrolizumab arm’s mortality hazard. Formally, $h_{\text{combo}}(t) = h_{\text{pembro}}(t)$ for $t > 60$ months, with the curve integrated forward from $S_{\text{combo}}(60)$, so no artificial survival discontinuity is introduced at month 60. The recurrence (r_{DM}) and local-recurrence (r_{LR}) ratio functions retain their base-case values, so any incremental QALY gain arises solely from the improved disease-state trajectory (longer recurrence-free and distant-metastasis-free intervals), scaled by the survival level already accrued during the observed period, rather than from any additional post-trial OS hazard benefit.

Table S15: Structural Scenario Analysis Results

Scenario	ΔLY	$\Delta QALY$	$\Delta Cost$	ICER	NMB \$100K	NMB \$150K
Base case	5.64	4.65	\$197,354	\$42,412	\$267,974	\$500,638
Weibull OS	8.34	6.47	\$368,253	\$56,911	\$278,818	\$602,354
Rank #1 joint model	8.34	6.44	\$393,705	\$61,128	\$250,362	\$572,395
Unconstrained (no GP)	9.33	7.28	\$376,788	\$51,740	\$351,440	\$715,554
GP floor 48 mo	5.63	4.64	\$196,847	\$42,395	\$267,468	\$499,626
GP floor 60 mo (base)	5.64	4.65	\$197,354	\$42,412	\$267,974	\$500,638
GP floor 72 mo	5.68	4.68	\$198,738	\$42,464	\$269,274	\$503,280
GP floor 96 mo	5.81	4.77	\$203,620	\$42,681	\$273,452	\$511,989
10-year horizon	1.31	1.31	\$96,674	\$73,791	\$34,337	\$99,842
20-year horizon	3.69	3.19	\$129,268	\$40,517	\$189,779	\$349,302
30-year horizon	5.12	4.27	\$175,562	\$41,159	\$250,989	\$464,265
40-year horizon (base)	5.64	4.65	\$197,354	\$42,412	\$267,974	\$500,638
Waning V1 (OS-survival convergence)	1.96	1.99	\$36,514	\$18,391	\$162,023	\$261,292
Waning V2 (hazard convergence)	2.92	2.30	\$240,389	\$104,590	−10,549	\$104,371
No additional OS hazard benefit beyond 5 years	1.81	1.86	\$38,751	\$20,824	\$147,337	\$240,381
0% discount	9.68	7.77	\$292,233	\$37,614	\$484,701	\$873,168
5% discount	4.11	3.46	\$168,485	\$48,726	\$177,294	\$350,183

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,740) but implausible \$376,788 cost.

Time horizon sensitivity. The ICER remained below conventional US willingness-to-pay thresholds across 20-, 30-, and 40-year horizons, although the point estimate increased at the 40-year horizon: \$40,517 at 20 years, \$41,159 at 30 years, and \$42,412 at the 40-year base-case horizon (Table S15). The 30-to-40-year increase of \$1,234/QALY (3.0%) reflects the limited contribution of late discounted QALY gains beyond 30 years; the 10-year horizon produced a higher ICER of \$73,791/QALY, reflecting the incomplete capture of long-term QALY gains within a shorter window. Because the general-population mortality hazard floor was applied from 60 months onward and both costs and QALYs were discounted at 3% annually, survival beyond the 20-year window contributed materially to discounted incremental costs and QALYs.

General population hazard floor start-time sensitivity. The GP mortality floor was applied from 60 months in the base case. Varying the onset from 48 to 96 months produced no material change in the ICER (the ICERs ranged from \$42,395 to \$42,681/QALY across 48–96 months). This reflects the fact that the fitted hazard for the combination arm closely tracks the general-population mortality hazard in the post-trial period (both at approximately 0.0008–0.0009 monthly), leaving the floor with minimal additional constraining effect regardless of onset timing. The floor’s binding is thus a near-numerical coincidence between the parametric extrapolation and US life-table mortality, not a model artifact.

10 Deterministic Threshold Acquisition Price

10.1 Price Threshold Analysis

Table S16: Price Threshold Analysis

Intismeran Price	ICER	NMB @ \$100K	NMB @ \$150K
\$0	Dominant	\$467,974	\$700,638
\$50,000	\$10,177	\$417,974	\$650,638
\$100,000	\$20,922	\$367,974	\$600,638
\$150,000	\$31,667	\$317,974	\$550,638
\$200,000 (base)	\$42,412	\$267,974	\$500,638
\$250,000	\$53,157	\$217,974	\$450,638
\$300,000	\$63,902	\$167,974	\$400,638
\$350,000	\$74,647	\$117,974	\$350,638
\$400,000	\$85,392	\$67,974	\$300,638
\$450,000	\$96,137	\$17,974	\$250,638
\$467,974	\$100,000	\$0	\$232,664
\$500,000	\$106,882	−32,026	\$200,638
\$550,000	\$117,628	−82,026	\$150,638
\$600,000	\$128,373	−132,026	\$100,638
\$650,000	\$139,118	−182,026	\$50,638
\$700,000	\$149,863	−232,026	\$638
\$700,638	\$150,000	−232,664	\$0
\$750,000	\$160,608	−282,026	−49,362
\$800,000	\$171,353	−332,026	−99,362

Deterministic threshold prices (from Table S16): approximately \$467,974 at \$100,000/QALY (2.3× base) and approximately \$700,638 at \$150,000/QALY (3.5× base). These are model-based deterministic estimates; PSA uncertainty propagation and formal budget-impact analysis would be needed to inform payer negotiations.

11 Reproducibility and Reporting

11.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at <https://github.com/yichao2022/cea-intismeran> (tag v1.0-submission).

Key dependencies: NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

Model details:

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: discrete annual, discount factor $(1 + r)^{-t}$ with $r = 0.03$ per year and t measured in years
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

Code structure:

- `model.py` — ModelParams, CEAModel (4-state PSM), discount_factors
- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, waning, no-post-trial-OS)
- `results_pipeline.py` — Canonical runner: base case, decomposition, scenarios, one-way DSA, price ladder, analytic thresholds, PSA delegation; enforces the consistency checks and exports the manuscript’s tables
- `psa_v2.py` — Probabilistic analysis with joint survival sampling (invoked by `results_pipeline.py --psa`)
- `survival_model_selection.py` — 15,625-combination model selection
- `figures.py`, `gen_tornado.py`, `gen_psa_output.py`, `gen_fig*.py` — Figure generation
- `output/survival_combos_15625.csv` — Full model selection results

Data extraction: Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

Reproducibility: Running `python3 results_pipeline.py` reproduces every reported quantity (base case, decomposition, scenarios, one-way DSA, price ladder and analytic threshold prices) from the single canonical model implementation, with algebraic consistency checks enforced before export; `python3 results_pipeline.py --psa` reproduces the probabilistic analysis. GitHub tag v1.0-submission freezes the submission code.

11.2 CHEERS 2022 Checklist

Table S17: CHEERS 2022 Checklist

No.	Item	Reported	Location
<i>Title</i>			
1	Title	Yes	Title page
<i>Abstract</i>			
2	Abstract	Yes	Structured abstract
<i>Introduction</i>			
3	Background and objectives	Yes	§1 (objectives §1.4)
<i>Methods</i>			
4	Health economic analysis plan	No	No analysis plan was developed for this academic evaluation of published data

No.	Item	Reported	Location
5	Study population	Yes	§2.2 (resected stage IIIB–IV melanoma; KEYNOTE-942, n=157)
6	Setting and location	Yes	§2.2, §2.4 (United States)
7	Comparators	Yes	§2.1, §2.2
8	Perspective	Yes	§2.4 (US payer)
9	Time horizon	Yes	§2.6 (lifetime, 40 years)
10	Discount rate	Yes	§2.6 (3% per year, both costs and outcomes)
11	Selection of outcomes	Yes	§2.1, §2.5 (life-years, QALYs)
12	Measurement of outcomes	Yes	§2.2 (digitized Kaplan–Meier), §2.3, §2.5
13	Valuation of outcomes	Yes	§2.5 (published utility values)
14	Measurement and valuation of resources and costs	Yes	§2.4
15	Currency, price date, and conversion	Yes	§2.4 (2025 US dollars; no currency conversion)
16	Rationale and description of model	Yes	§2.1 (four-state partitioned survival model); model code, title page
17	Analytics and assumptions	Yes	§2.2, §2.3, §2.6, §2.7
18	Characterizing heterogeneity	Not reported	No subgroup analyses were performed
19	Characterizing distributional effects	Not reported	Distributional effects were not modelled
20	Characterizing uncertainty	Yes	§2.7; Supplement §8–9
21	Approach to engagement with patients and others affected by the study	Not applicable	No patient, public or stakeholder engagement was undertaken
<i>Results</i>			
22	Study parameters	Yes	§2.2–2.5; Supplement §6 (all analytic inputs and ranges)
23	Summary of main results	Yes	§3.1, §3.2 (Tables 2–3)
24	Effect of uncertainty	Yes	§3.4–3.6 (DSA, PSA, structural scenarios, price thresholds); Supplement §8–9
25	Effect of engagement with patients and others affected by the study	Not applicable	No engagement was undertaken
<i>Discussion</i>			
26	Study findings, limitations, generalizability, and current knowledge	Yes	§4.1, §4.5, §4.6
<i>Other relevant information</i>			
27	Source of funding	Yes	Title page
28	Conflicts of interest	Yes	Title page

Reported, Not applicable and Not reported are used as directed by the CHEERS 2022 statement [11]. Locations refer to the manuscript unless a supplement section is named. No analysis plan was registered or published for this evaluation, no subgroup or distributional analyses were undertaken, and no patients or stakeholders were engaged, as this is a model-based evaluation of published aggregate trial data.

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